

Kramers nonlinearities in PT symmetric magnets

Oles Matsyshyn, Y Xiong, J Song,

arXiv 2406.19444 (2024),

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IPS Meeting 2024
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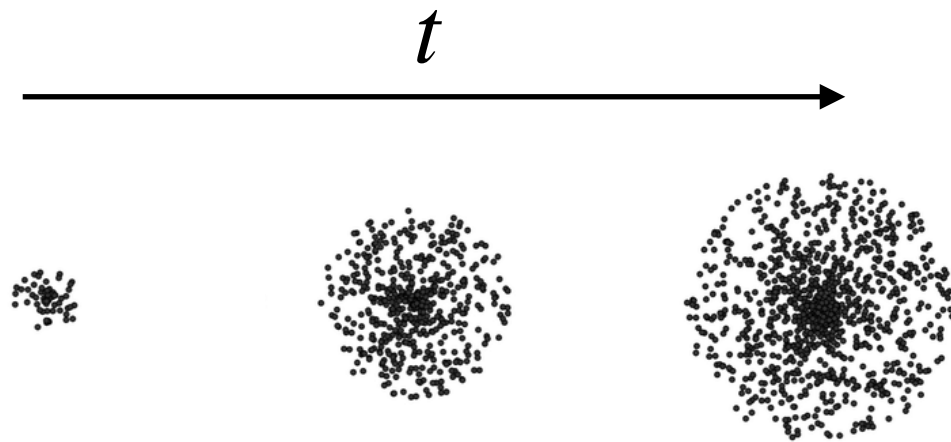
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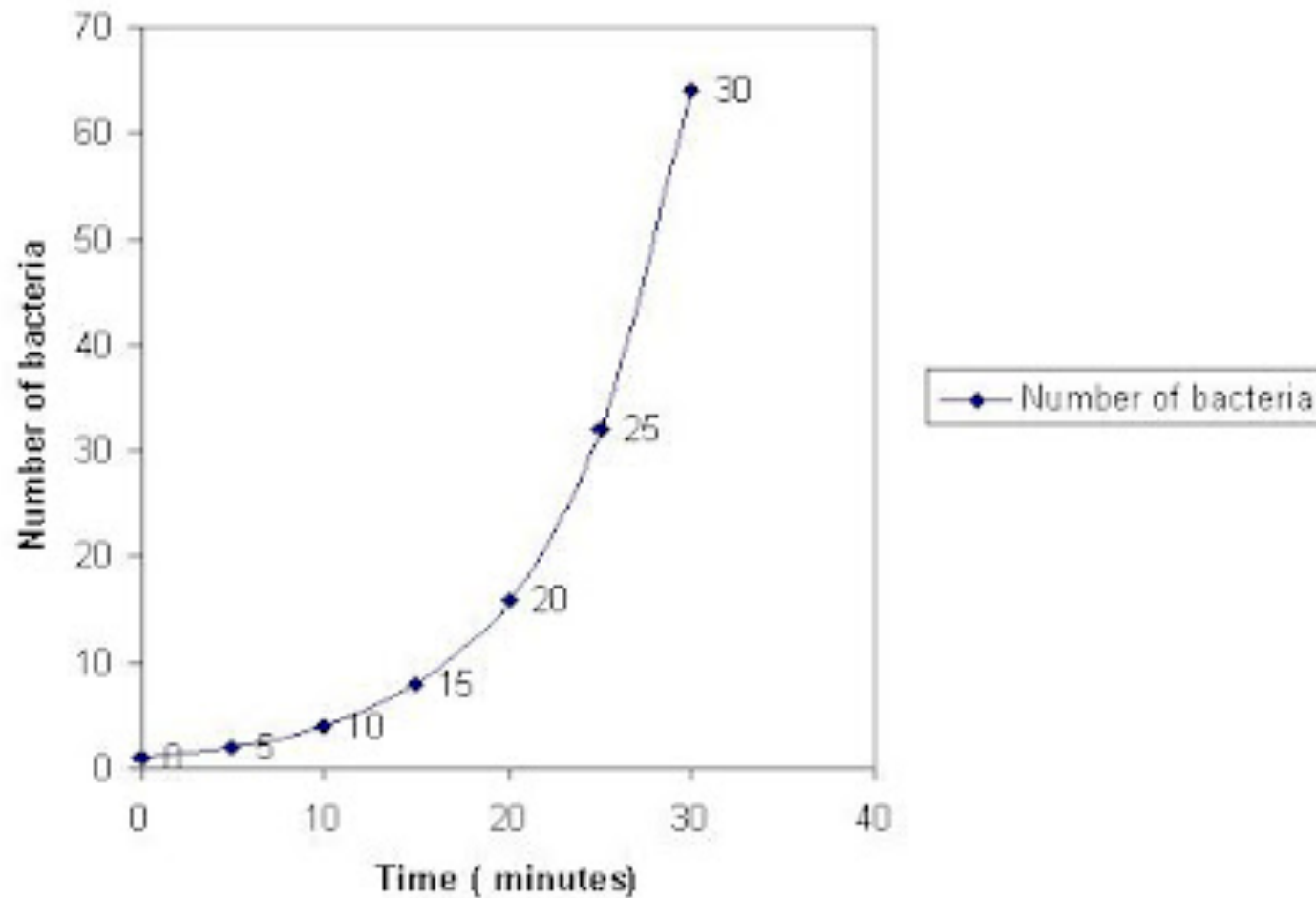


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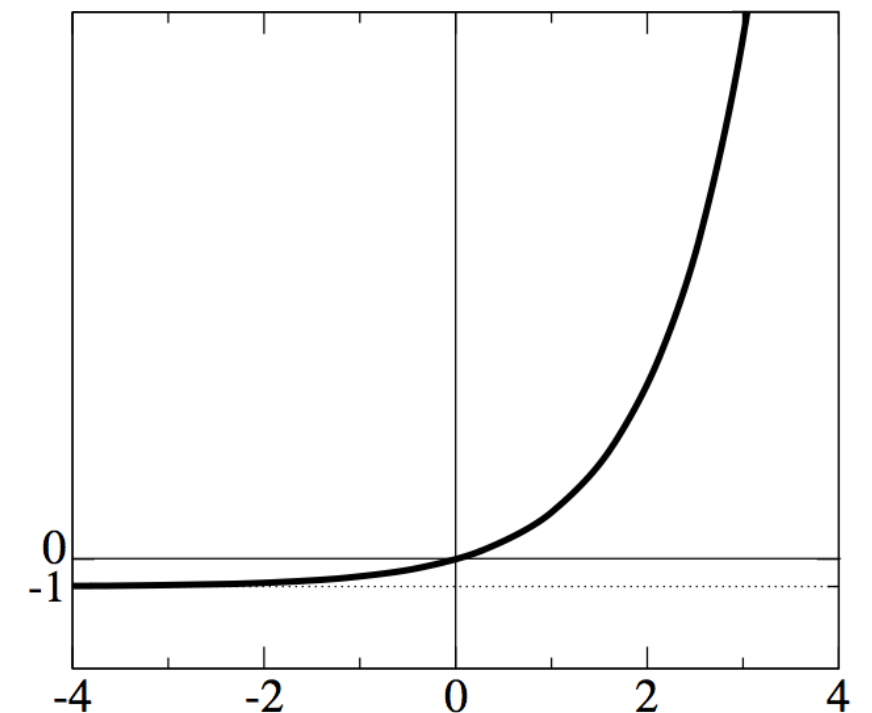
Classic nonlinearities



Bacterial population growth



For non-reciprocal devices



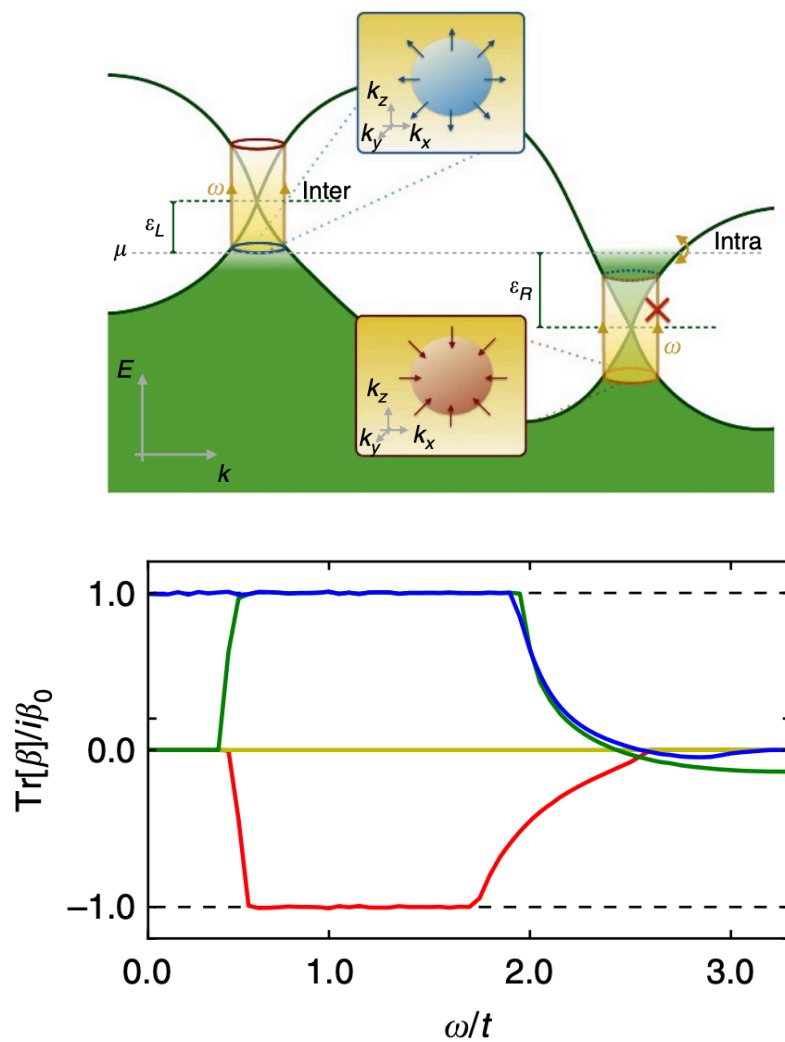
backward

forward

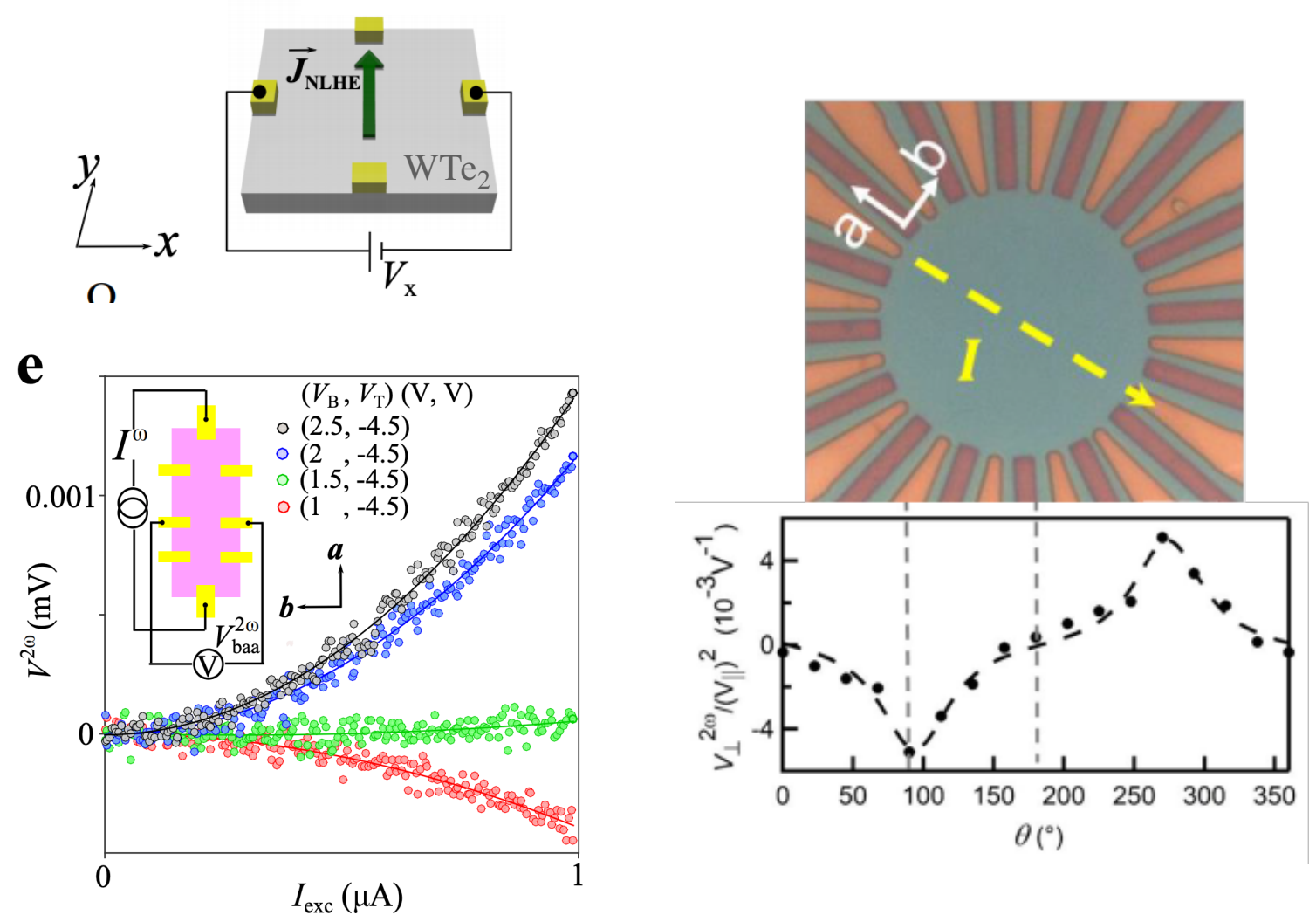
Can quantum materials host a special nonlinearities?

Nonlinear photocurrent responses

Quantized Circular photogalvanic effect in Weyl semimetals



Nonlinear Hall response in a TRS preserving system



De Juan, et al., Nature Communications 8, 15995 (2017)

Ma, et al., Nature 565, 337-342 (2019)
similar work found in Kang et al., Nature Materials 18, 324-328 (2019)

Diversity of second-order nonlinear photocurrents

“Metallic” photocurrents

so-called “Fermi surface” contributions

e.g.,

(i) Berry curvature dipole nonlinear transport

arises from (berry curvature induced)
anomalous velocity contributions to
the semiclassical equations of motion
(in a band)

e.g., TRS: Orenstein and Moore PRL (2010), Sodemann and Fu PRL (2015)
PT: Gao, Yang, Niu (2014), Wang, Gao, Xiao PRL (2020),

(ii) Extrinsic scattering

e.g., from skew scattering-type processes

e.g., TRS: Du, ..., Xie Nat Comm (2019), Konig, ..., Pesin PRB (2019),
Isobe, Xu, L Fu Sci Adv (2020),
PT: M Da, A Arora, G Vignale, JS PRL (2023)

(iii) Intrinsic Fermi surface effect

arising from off-resonant contributions
in the second order density matrix

e.g., de Juan, Grushin PRR (2020), Wantanabe and Yanase PRX
(2021), Gao, ... Rappe PRR (2021),

“Insulator” photocurrents

Real transitions between two bands

(i) Shift Photocurrent

arises from “shift” in the quasiparticle’s
center of mass when it transitions
from valence to conduction band

e.g., von Blatz and Kraut PRB (1981), Morimoto and
Nagaosa, Sci. Adv (2016)....

(ii) Injection Photocurrent

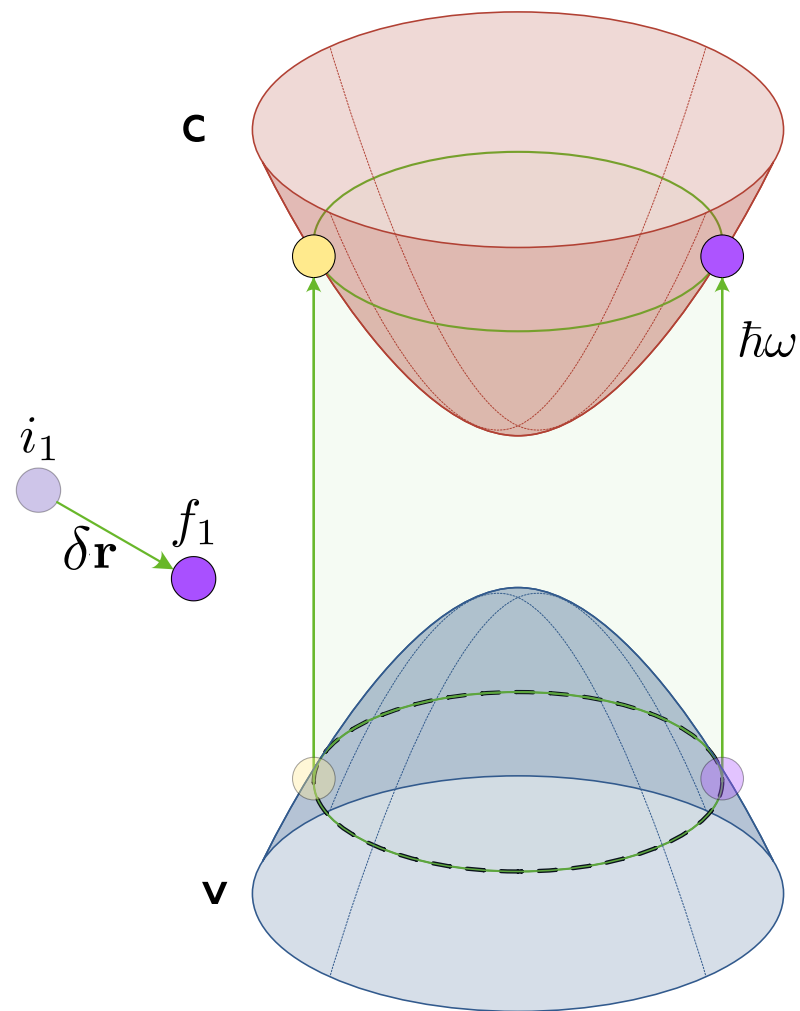
arises from difference in a particle’s
velocity when it transitions from
valence to conduction band

e.g., Sipe PRB (2000), De Juan et al., Nat Comm (2016)....

Non-exhaustive list: Other types of bulk photocurrents can
also be induced, these are just a sampling of some of the
most well-known. Lots of other papers other than the ones
shown here too; these are just examples.

Two prominent photocurrents: Shift and Injection

Shift photocurrent

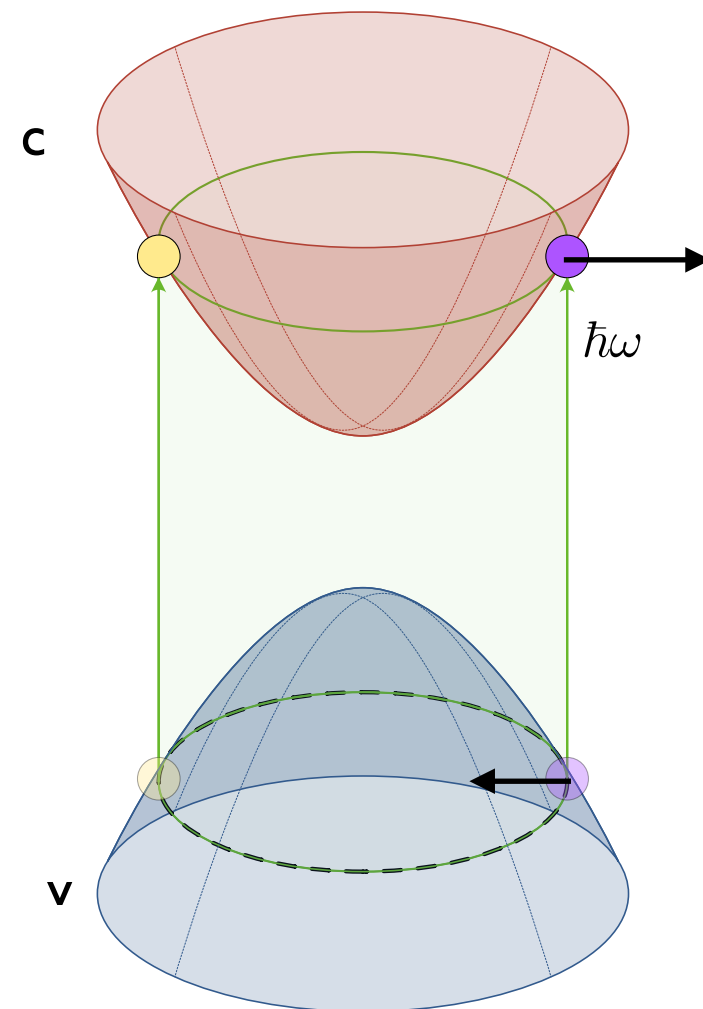


shift vector = shift in
wavepacket centre

$$\mathbf{j}^s = e \sum_{i \rightarrow f} W_{i \rightarrow f} \delta \mathbf{r}_{i \rightarrow f},$$

↑
Interband transition rate

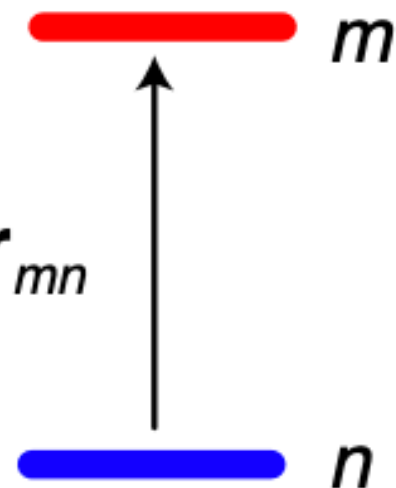
Injection photocurrent



velocity difference
 $\Delta_{i \rightarrow f} = \mathbf{v}_f - \mathbf{v}_i,$

$$\partial_t \mathbf{j}^{\text{inj}} = e \sum_{i \rightarrow f} W_{i \rightarrow f} \Delta_{i \rightarrow f},$$

Quantum Geometry of optical transitions

$$\langle \psi_m | \hat{H}_{\text{int}} | \psi_n \rangle = e \mathbf{E} \cdot \mathbf{r}_{mn}$$


The diagram shows two horizontal bars representing energy levels. The lower bar is blue and labeled 'n'. The upper bar is red and labeled 'm'. A vertical arrow points from the blue bar to the red bar, indicating a transition between the two states.

geometrical structure depends on inner product

$$(A, B) = \sum_{\alpha, \beta} A_{\alpha\beta}^* B_{\alpha\beta} = \text{Tr} [A^\dagger B].$$

for interband optical transitions:

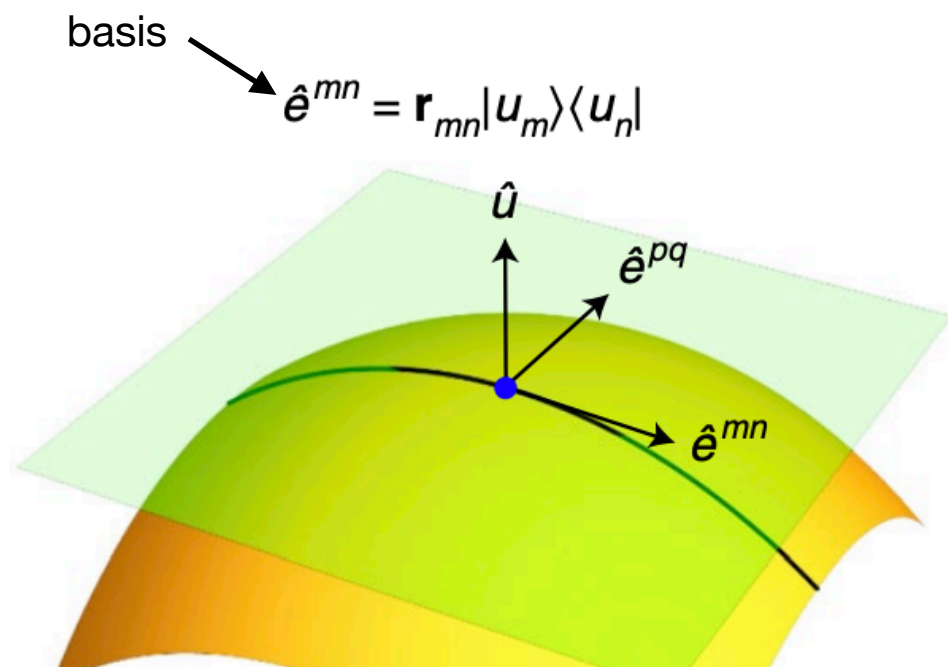
$$Q_{ba}^{mn} \equiv (\hat{e}_b^{mn}, \hat{e}_a^{mn}) = r_{nm}^b r_{mn}^a.$$

parallel transport of how vectors change with k

$$\nabla_c \hat{e}_a^{mn} = \sum_b (C^{mn})_{ca}^b e_b.$$

connection

$$C_{bca}^{mn} = (\hat{e}_b^{mn}, \nabla_c \hat{e}_a^{mn}) = r_{nm}^b r_{mn,c}^a,$$



Quantum geometry and photocurrent

Geometrical meaning of shift currents

Hermitian Connection

$$\sigma_{\text{shift}}^{c;ab} = -\frac{\pi e^3}{2\hbar^2} \sum_{m,n} \int_{\mathbf{k}} \delta_{\epsilon} f_{nm} i(C_{bca}^{mn} - (C_{acb}^{mn})^*).$$

Ahn, Guo, Nagaosa, Vishnawath, Nature Physics 18, 290-295 (2022)

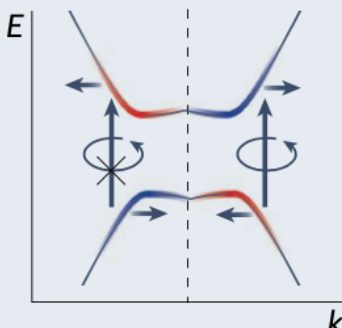
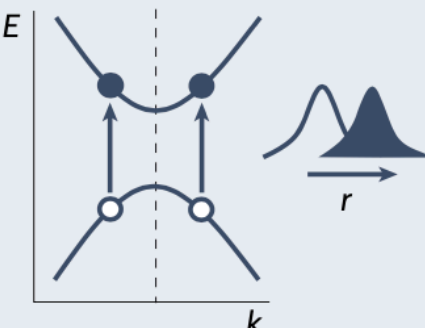
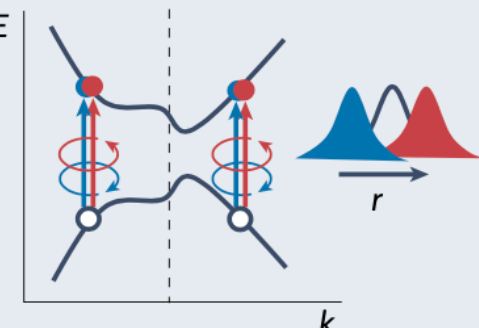
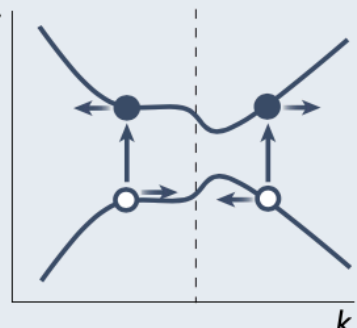
Geometrical meaning of injection currents

Quantum geometric tensor

$$\sigma_{\text{inj}}^{c;ab} = -\frac{\pi e^3}{\hbar^2 \Gamma} \sum_{m,n} \int_{\mathbf{k}} \delta_{\epsilon} f_{nm} Q_{ba}^{mn} (v_{mm}^c - v_{nn}^c),$$

many people: e.g., Ahn, Guo, Nagaosa, Phys Rev X 10, 041041 (2020)
(some term this a quantum geometric “dipole”)

other “geometrical” interpretations from
photocurrent: e.g., Binghai Yan,

Charge photocurrent	Circular injection	Linear shift	Circular shift	Linear injection
<i>P</i> symmetry	×	×	×	×
<i>T</i> symmetry	✓	✓	×	×
<i>PT</i> symmetry	×	×	✓	✓
Material examples	TaAs, CoSi, RhSi, Td-W(Mo)Te ₂	TaAs, CoSi, RhSi, Td-W(Mo)Te ₂	Even-layered MnBi ₂ Te ₄ Even-layered CrI ₃	Even-layered MnBi ₂ Te ₄ Even-layered CrI ₃
Microscopy process				

What happens in PT symmetric layered materials?

Injection nonlinearity in non-degenerate bands

injection nonlinearity for an observable O

(e.g., O = current, layer polarization,)

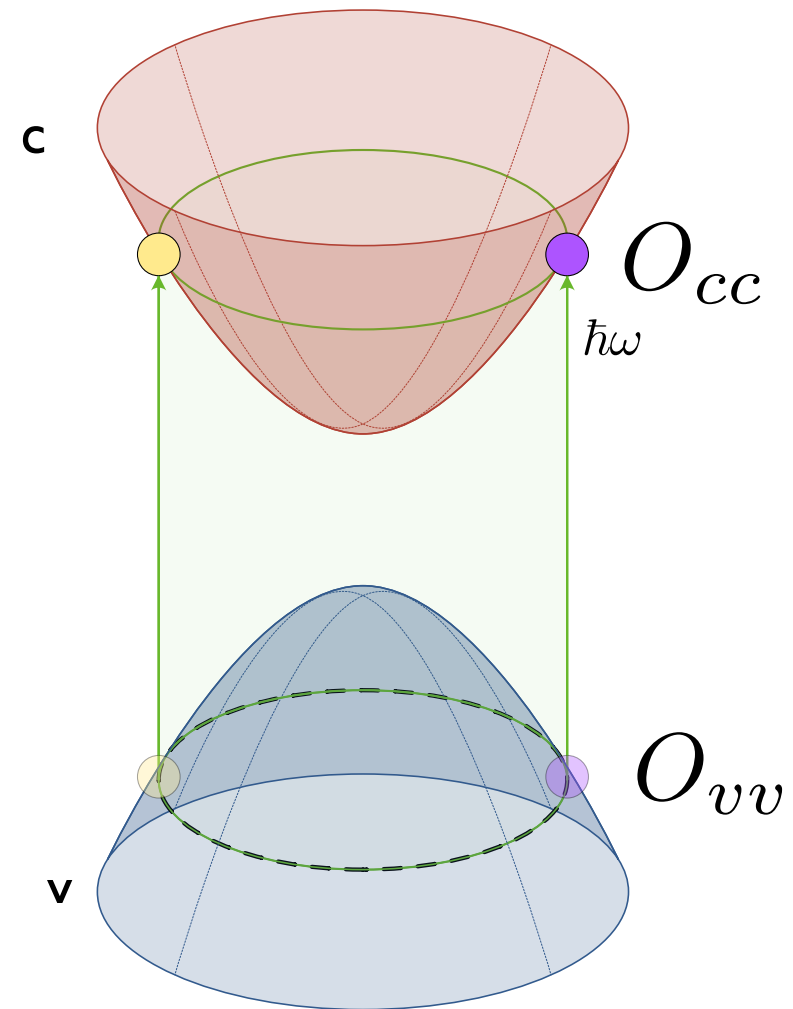
$$\frac{dO}{dt} = \sum_{v \rightarrow c} [O_{cc}(\mathbf{p}) - O_{vv}(\mathbf{p})] f_{cv}(\mathbf{p}) W_{v \rightarrow c}(\mathbf{p}),$$

matrix element $O_{nn}(\mathbf{p}) = \langle n(\mathbf{p}) | \hat{O} | n(\mathbf{p}) \rangle$

transition rate $W_{v \rightarrow c}(\mathbf{p}) \propto r_{vc}^{\alpha}(\mathbf{p}) r_{cv}^{\beta}(\mathbf{p})$

process involves 2 states

$$\begin{array}{c} c \quad \overline{O_{cc}} \\ \begin{array}{c} r_{vc} \quad | \quad r_{cv} \\ | \quad | \\ \hline \end{array} \\ v \quad \overline{O_{vv}} \end{array}$$



Injection nonlinearity in non-degenerate PT bands

injection nonlinearity for an observable O

(e.g., O = current, layer polarization,)

$$\frac{dO}{dt} = \sum_{v \rightarrow c} [O_{cc}(\mathbf{p}) - O_{vv}(\mathbf{p})] f_{cv}(\mathbf{p}) W_{v \rightarrow c}(\mathbf{p}),$$

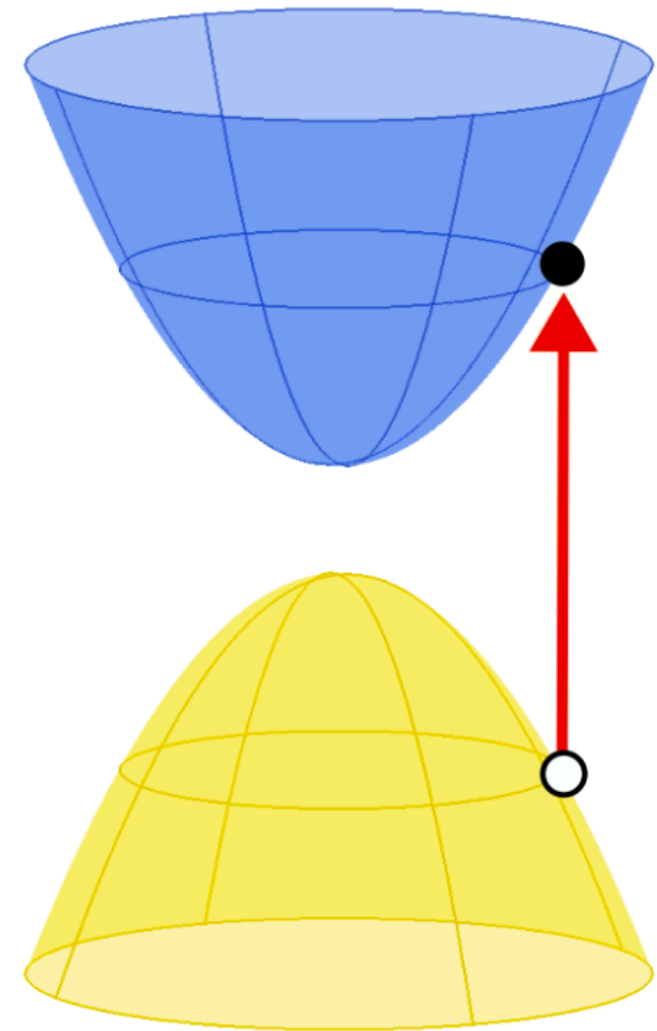
where $O_{nn}(\mathbf{p}) = \langle n(\mathbf{p}) | \hat{O} | n(\mathbf{p}) \rangle$

Case I: O = current = ev

Produces familiar injection photocurrent

Observable is PT-even

expectation value is non-vanishing in each Bloch state



Injection nonlinearity in non-degenerate PT bands

injection nonlinearity for an observable O

(e.g., O = current, layer polarization,)

$$\frac{dO}{dt} = \sum_{v \rightarrow c} [O_{cc}(\mathbf{p}) - O_{vv}(\mathbf{p})] f_{cv}(\mathbf{p}) W_{v \rightarrow c}(\mathbf{p}),$$

where $O_{nn}(\mathbf{p}) = \langle n(\mathbf{p}) | \hat{O} | n(\mathbf{p}) \rangle$

Case 1: O = current = ev

Produces familiar injection photocurrent

Observable is PT-even

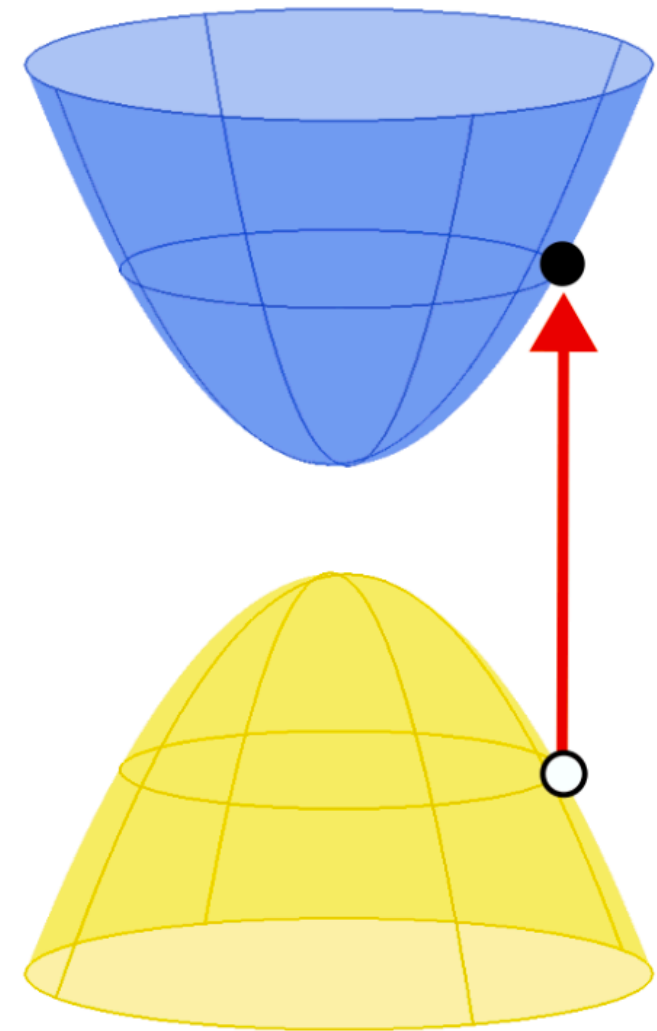
expectation value is non-vanishing in each Bloch state

Case 2: O = polarization/magnetization

Observable is PT-odd

Expectation value in each Bloch state **vanishes!**

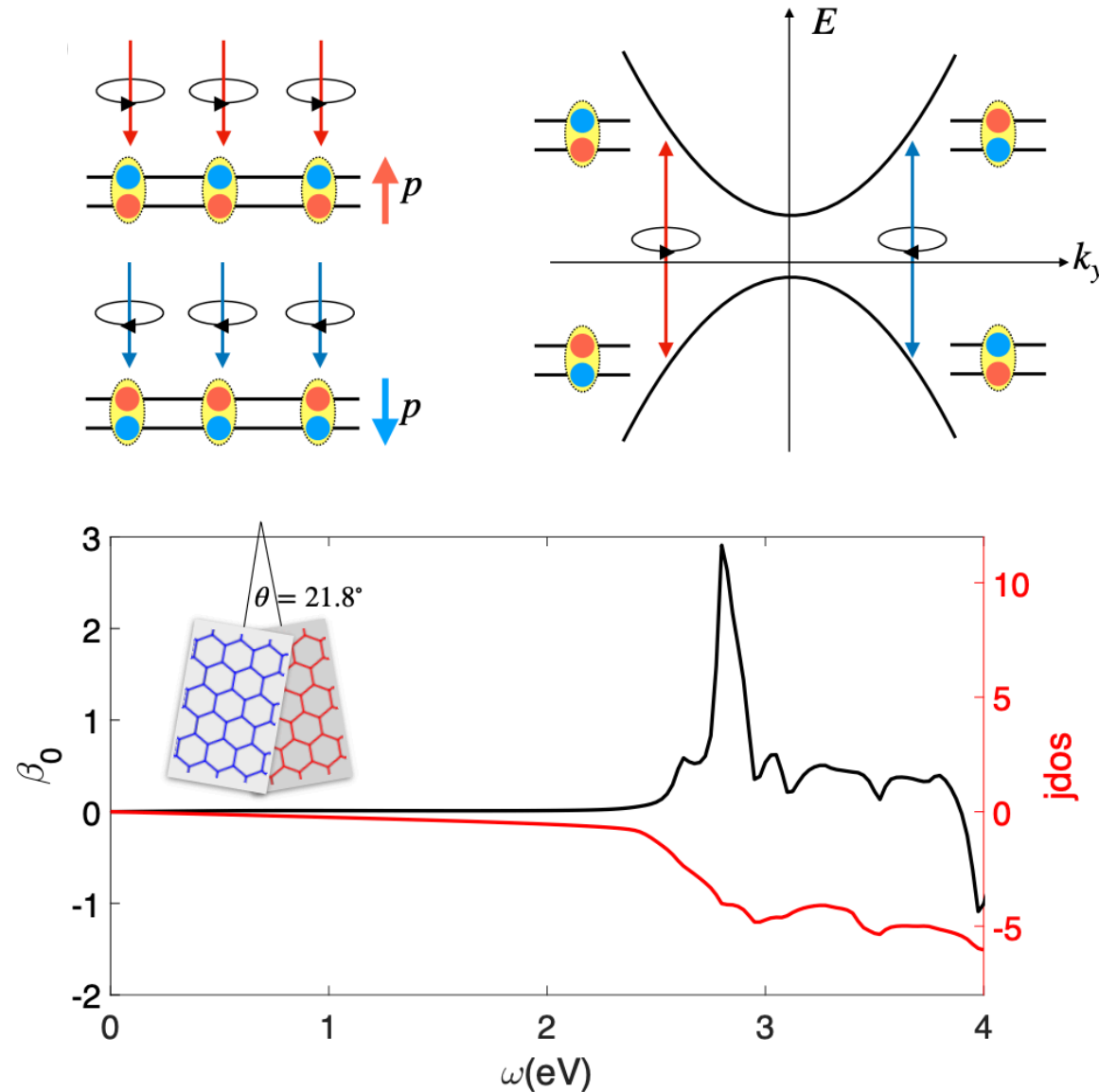
expect this for all PT-odd observables: polarization, magnetization, ...



Nonlinear layer polarization responses

Shift type

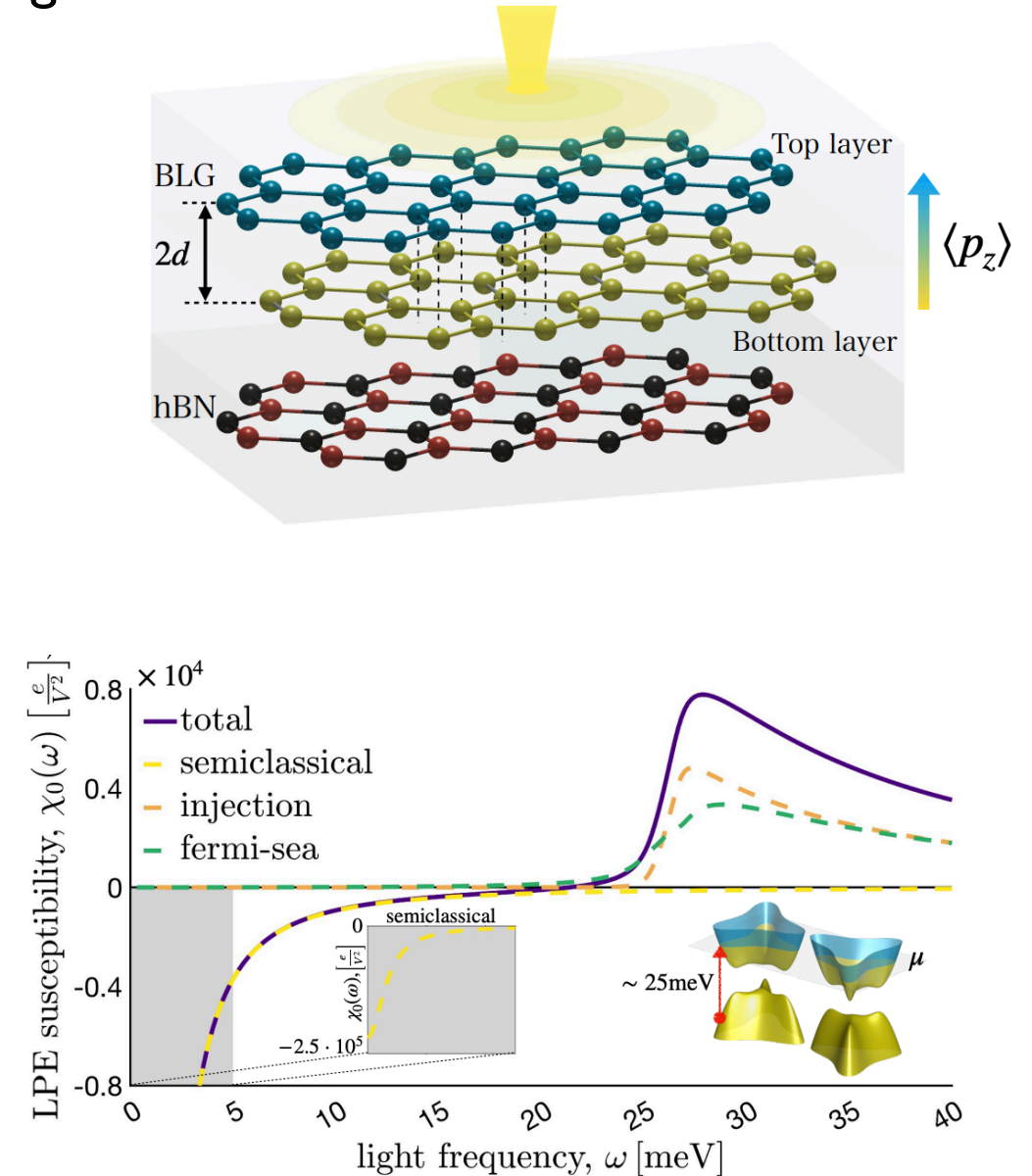
Layer polarization induced by circularly polarized light in chiral bilayers (e.g., TBG)



Y Gao, Y Zhang, D Xiao PRL 124, 077401 (2021)

Injection type

Layer polarization induced by unpolarized light



O Matsyshyn, Y Xiong, A Arora, JS, PRB 107, 205306 (2023)

* there are also “Fermi sea” layer polarization responses. See O Matsyshyn, Y Xiong, A Arora, JS, PRB 107, 205306 (2023)
see also H Zheng, D Zhu, C Xiao, W Yao, Nanoletters 24, 8017 (2024); See Matsyshyn et al, for Fermi surface responses

Kramers degeneracy in PT symmetric materials

Kramers degeneracy in PT symmetric materials:

every $\mathbf{p}$ has two orthogonal states

$$|u_n(\mathbf{p})\rangle \text{ and } |u_{\bar{n}}(\mathbf{p})\rangle$$

that are degenerate in energy

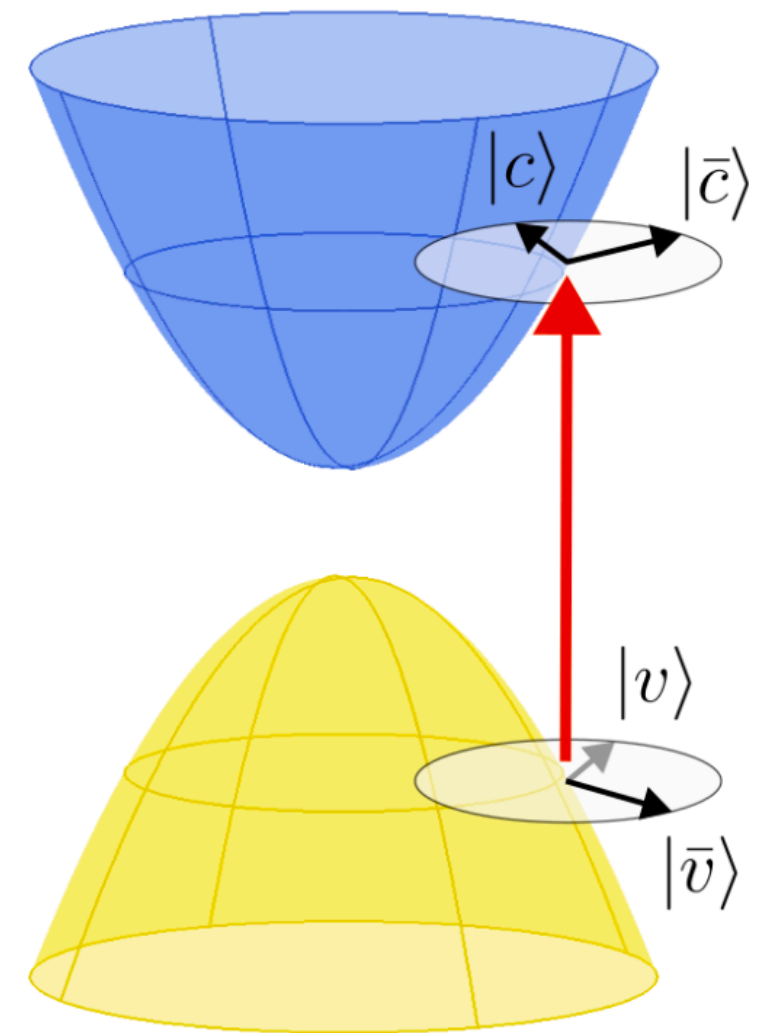
$$\epsilon_n(\mathbf{p}) = \epsilon_{\bar{n}}(\mathbf{p})$$

Observables are also constrained

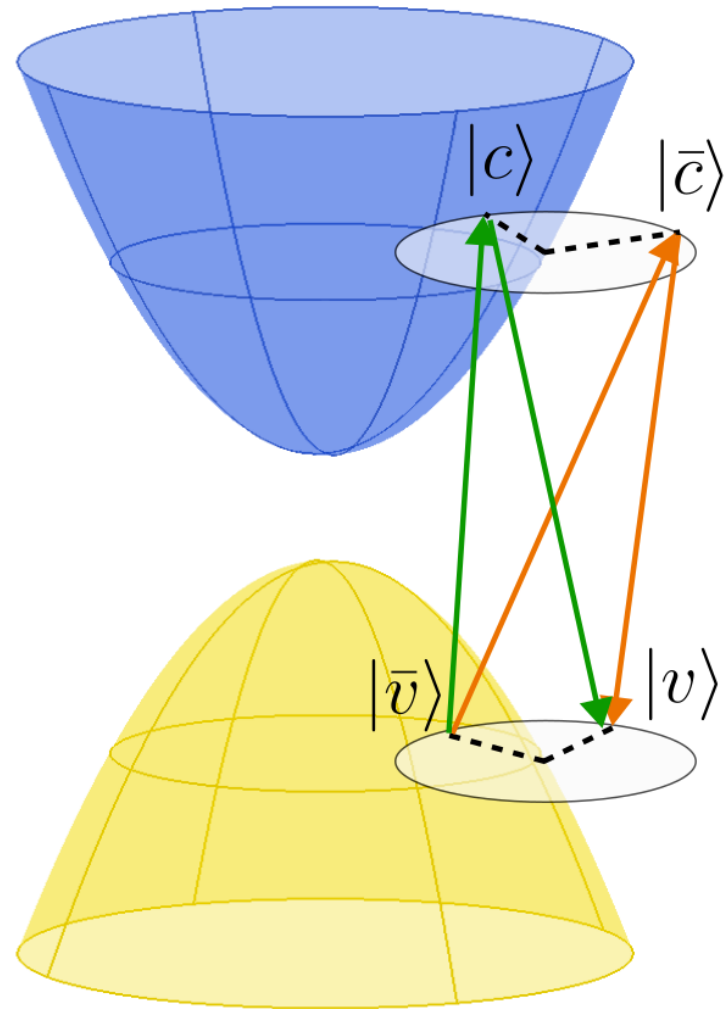
$$O_{nn}^{(\pm)}(\mathbf{p}) = \pm O_{\bar{n}\bar{n}}^{(\pm)}(\mathbf{p}), \quad O_{n\bar{n}}^{(\pm)}(\mathbf{p}) = \mp O_{\bar{n}n}^{(\pm)}(\mathbf{p}),$$



expectation value for PT odd no longer vanishes identically!



Injection nonlinearity in degenerate PT bands



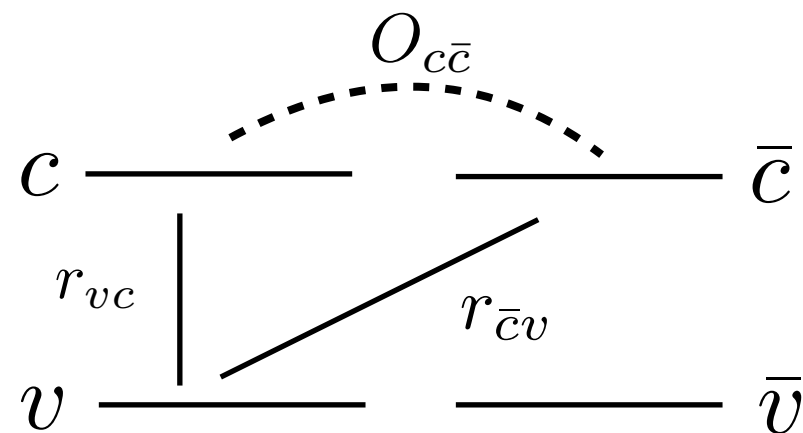
injection nonlinearity for an observable O

$$\frac{dO}{dt} = \frac{\pi e^2}{2\hbar} \sum_{\substack{mn \\ \zeta=\pm 1}} \delta(\epsilon_{mn} - \zeta \hbar \omega) r_{nm}^\alpha \mathbb{O}_{mn}^\beta f_{nm} E_{-\zeta\omega}^\alpha E_{\zeta\omega}^\beta,$$

where

$$\mathbb{O}_{mn}^\beta = \sum_{i, \epsilon_i = \epsilon_m} O_{mi} r_{in}^\beta - \sum_{j, \epsilon_j = \epsilon_n} O_{jn} r_{mj}^\beta$$

sums across degenerate states (3 state process)

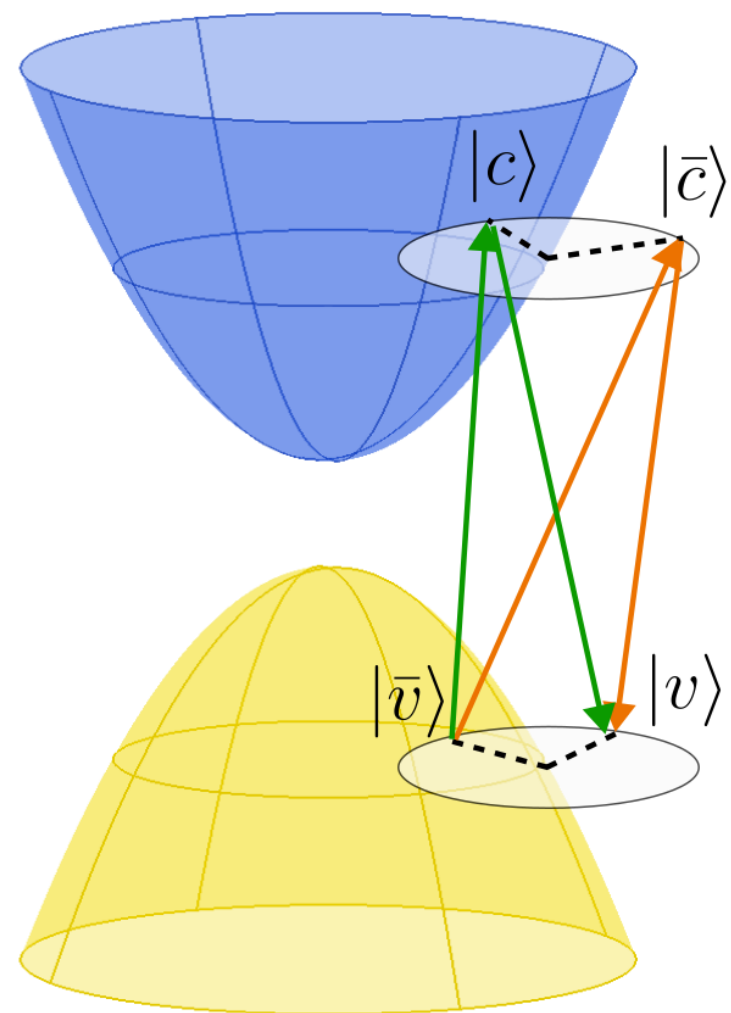


PT symmetry constraints

$$O_{nn}^{(\pm)}(\mathbf{p}) = \pm O_{\bar{n}\bar{n}}^{(\pm)}(\mathbf{p}),$$

$$O_{n\bar{n}}^{(\pm)}(\mathbf{p}) = \mp O_{\bar{n}n}^{(\pm)}(\mathbf{p}),$$

Kramers nonlinearities

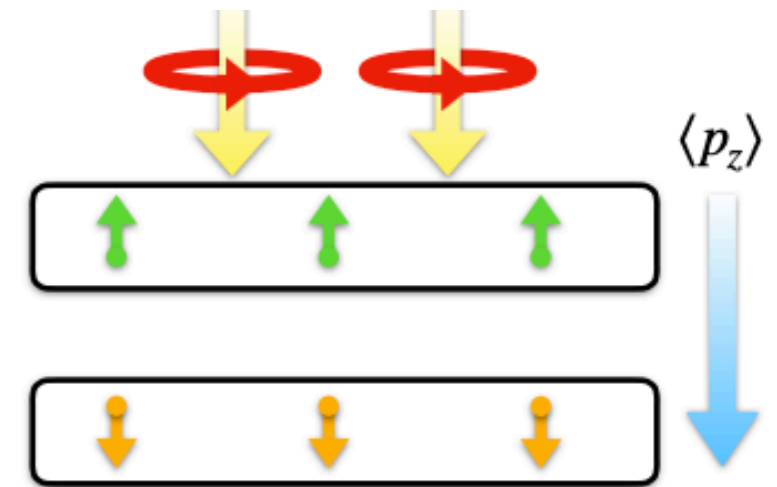
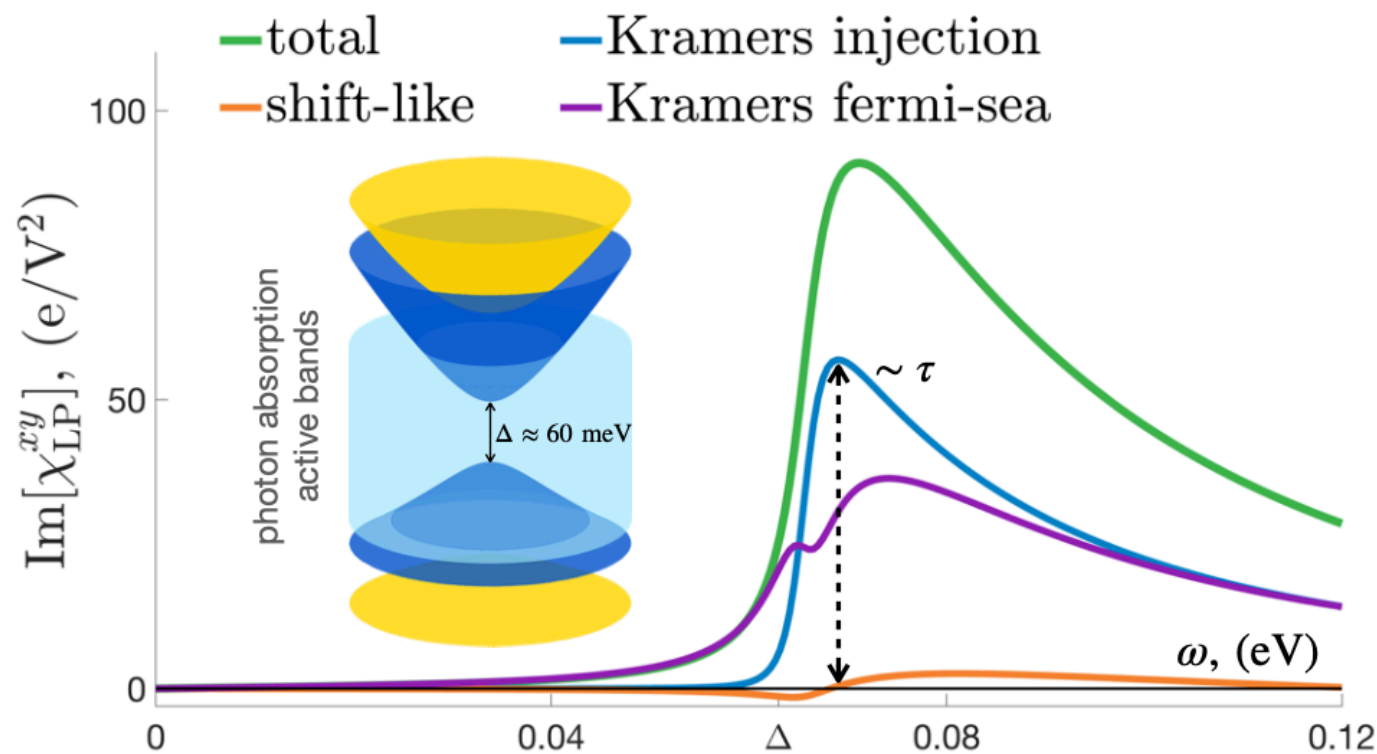


PT	Degenerate	Non-Degenerate
Kramers Injection	✓	✗
Kramers Fermi Sea	✓	✗
Shift – like	✓*	✓

- Key Properties:
1. Vanishes for non-degenerate PT bands
 2. Manifests for PT odd observables (polarization)
 3. Cannot be decomposed into non-degenerate formula

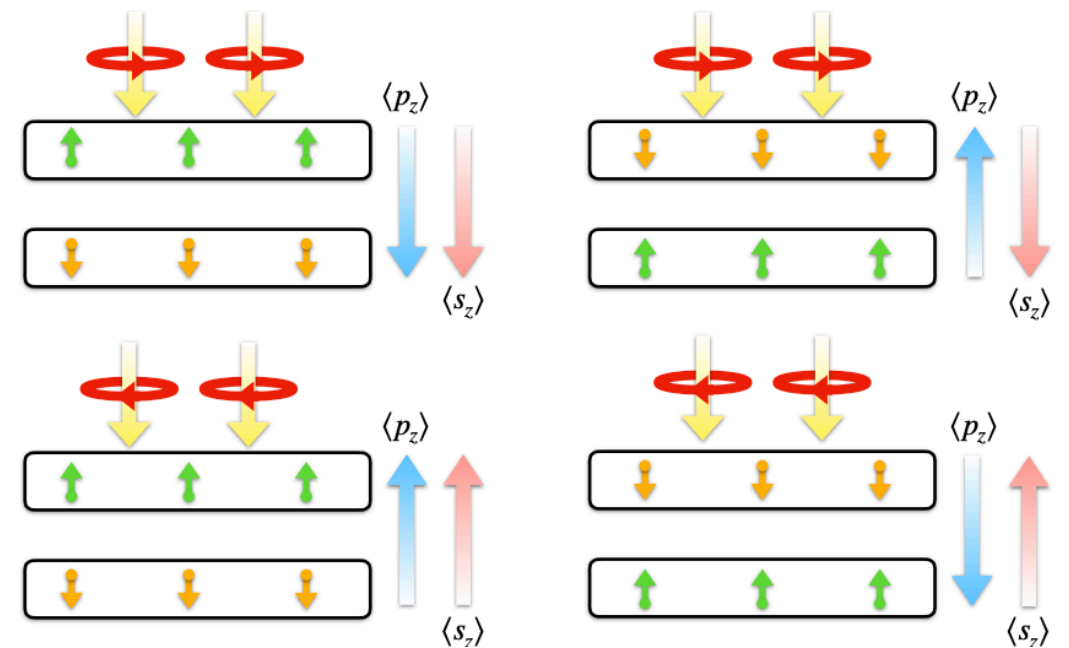
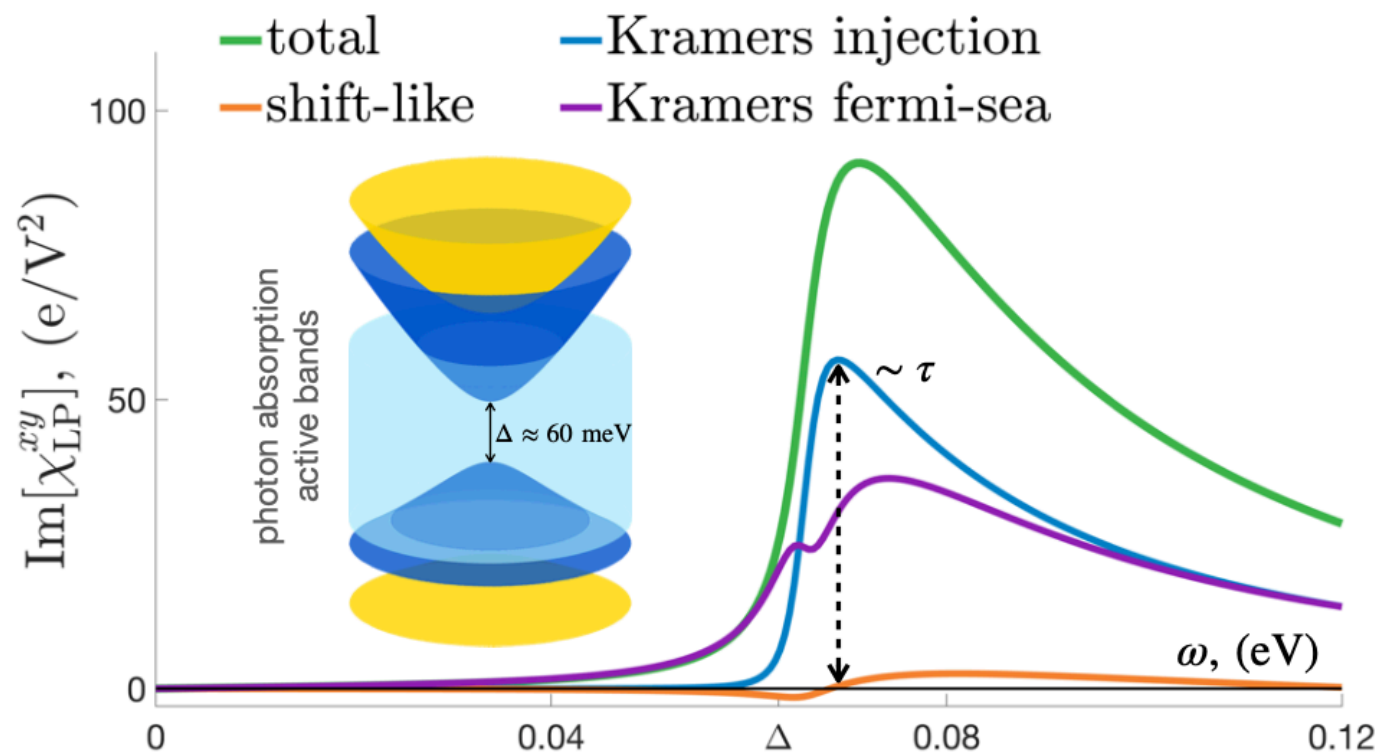
Kramers nonlinearities in MBT

Kramers nonlinear layer polarization susceptibility in MBT

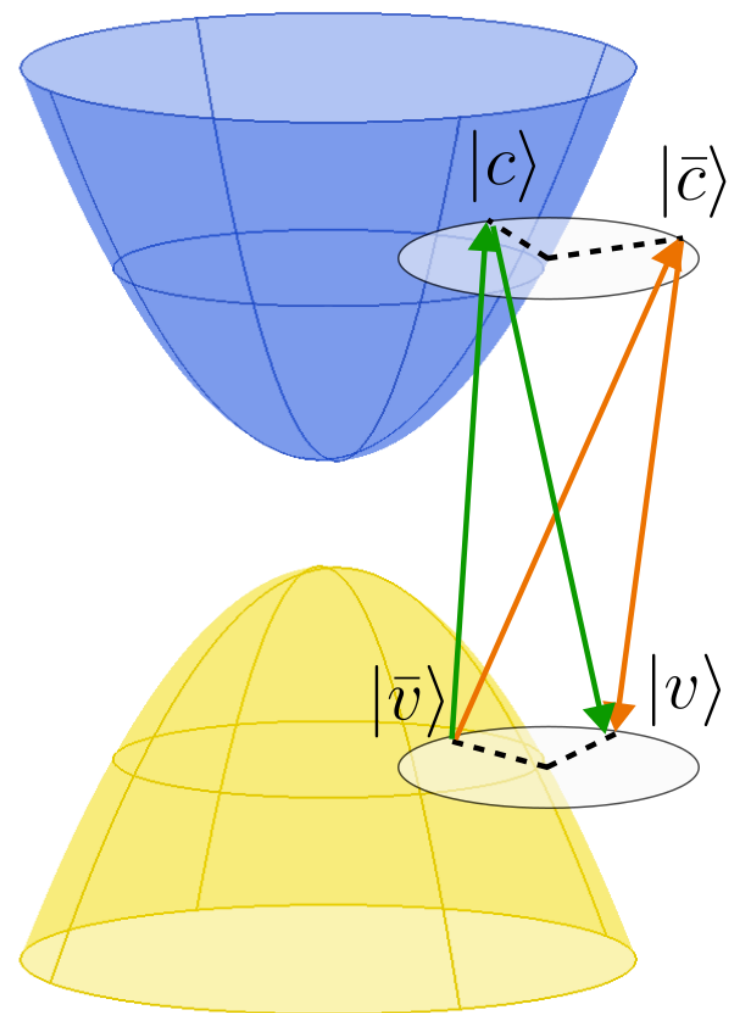


Kramers nonlinear layer polarization in MBT

Kramers nonlinear layer polarization susceptibility in MBT



Kramers nonlinearities



PT	Degenerate	Non-Degenerate
Kramers Injection	✓	✗
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Shift – like	✓*	✓

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